

ENGINEER ROOTED IN AUTOMATION, MACHINE SOFTWARE, AND INDUSTRIAL SYSTEMS WITH EXPERIENCE ACROSS MANUFACTURING, ROBOTICS AND EMBEDDED TECHNOLOGIES. I BRING SYSTEM-LEVEL THINKING, I VALUE CLARITY AND CONTINUOUS IMPROVEMENT. I AM SEEKING ROLES THAT OFFERS STRONGER OWNERSHIP AND OPPORTUNITY TO DESIGN, BUILD OR ARCHITECT IMPACTFUL ENGINEERING SOLUTIONS - WHETHER IN SOFTWARE, ROBOTICS OR SYSTEMS-ARCHITECTURE.

## EDUCATION

|                                                   |        |
|---------------------------------------------------|--------|
| MS, ELECTRICAL & COMPUTER ENGINEERING (3.41/4)    | DEC-17 |
| TANDON SCHOOL OF ENGINEERING, NEW YORK UNIVERSITY |        |
| BE, INSTRUMENTATION & CONTROLS, (7.04/10)         | JUN-14 |
| SVIT, GUJARAT TECHNOLOGICAL UNIVERSITY            |        |

## SKILLS

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| COMPUTER PROGRAMMING: | C/C++/CS, JAVA (DESKTOP & WEB), GIT, PYTHON (AI-AUTOMATION TOOLS, LIMITED HANDS-ON CODING), WEB(HTML, CSS, JS), DESKTOP(JAVAFX, AWT) |
| AUTOMATION:           | MICRO-CONTROLLERS & SINGLE BOARD COMPUTERS, PLCs(LADDER-LOGIC, FBD), HMI/SCADA                                                       |
| INDUSTRIES SERVED:    | MACHINERIES, MANUFACTURING, CONTROL PANELS, SOLAR ENERGY, EDUCATION                                                                  |

## PROFESSIONAL EXPERIENCE

### SOFTWARE ENGINEER III - AUTOMATION, UNIVERSAL INSTRUMENTS, NY Sep 22 - present

- Developing and maintaining **Fuzion** platform software on regular basis.
- Engineered **SBS (Side Barcode Scanning)** using Keyence SR-X300 with Live View, traceability, error recovery and multi-bank support for Connectors with barcodes on side of the body.
- Upgraded existing **AMS** process from single point to 13-locations for higher accuracy on wafer.
- Built a **wafer-to-board mapping** system that converts customer wafer map formats into Universal Fuzion board template files for direct use in production.
- Developed **Selective Placement** with parts buffering, DFF-based Patch & Interposer categories, and avoiding-priority rules for optimal patch-interposer matching.
- Implemented **AVP** to allow user-defined fiducial finds during ECS, NPI and Production flows.
- Integrated Cognex QuickBuild (VPP) with **MCOS↔QBIT** TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Fuzion Precision-Bonder platform.
- Built a CS **Light Intensity Controller** software with multi-channel control, network connectivity and controller discovery for illumination management.
- Implemented **nested-tray** support for Multi-Rot feeders.
- Added **isosceles triangle** fiducial support.
- Implemented configurable **park-position** on Fuzion platform.
- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree.

### AUTOMATION ENGINEER , NOR-CAL CONTROLS, CA Jan - Jul 22

- Developed automation software for utility-scale solar power plants
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

### SOFTWARE ENGINEER - MACHINE AUTOMATION, MINI-CIRCUITS, NY Feb 18 - Nov 21

- Developed and maintained software for pick-and-place rotary machines with computer vision inspection.
- Implemented Reverse motion on rotary machines with Pressing-arm to enable EVT testing.
- Implemented the new shopping cart system for the online store <https://www.minicircuits.com/>
- Developed enterprise application software for internal engineering teams.
- Tech stack: Java · JavaFX · HTML · CSS · JavaScript

### PRIMA AUTOMATION, AHMEDABAD, INDIA Jun 14 - Jan 16

### DESIGN ENGINEER May 15 - Jan 16

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.

### QUALITY CONTROL ENGINEER Jun 14 - May 15

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.

## ACADEMIC WORK EXPERIENCE

### DEVELOPER INTERN, REVMAX FLEET OPTIMIZATION, NY Sep - Dec 17

- Developed firmware for the SidewalkIQ sensor using Doppler radar to monitor sidewalk activity.
- Deployed sensors at multiple NYC locations and validated field performance
- Conducted R&D to improve data consistency and optimize storage and transfer to backend servers.
- Tech stack : · Arduino-C ·

### INSTRUCTOR (ITEST PROGRAM), ROBOTICS & MECHATRONICS DPT., NYU Sep - Dec 17

- Mentored to Robotics Club Programming Team for competitions (First Robotics & Vex Robotics).
- Taught concepts of Computer Programming to Robotics Club Team.

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
  - Autonomous Car (Mouse in a Maze) - STM32 Board: [https://github.com/sap586/RTES\\_Project](https://github.com/sap586/RTES_Project)
  - Wireless Robot Controller - Arduino: <https://github.com/sap586/WirelessRobotDay5Activity>